Reg. No. | | | | | | | | | | |

[18CSI302]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2021

OBJECT ORIENTED PROGRAMMING USING JAVA

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. Describe 5 features of Java. Write down the steps to execute a java program from command line. **07**

b. Outline the break, continue, and return keywords with sample code snippet. **07**

2. a. Explain method Overloading. Write two overloaded methods which takes 2 parameters but in different order. The method adds the values passed and add them and print them. **07**

b. Define Abstract class and method. Construct a java program with an abstract class named "Animal" which has an abstract method named "animalSound()". The abstract method is overridden by a class which inherited Animal class. Create object of the subclass and call the overridden method. **07**

3. a. Classify String and StringBuffer with 5 differences. Provide signature of 3 String constructors. **08**

b. Consider the string "Learn Share Learn". Write a program to explain the following methods and their differences in searching out the following strings "ea": indexOf(String), indexOf(String,int) and lastIndexOf(String). **06**

OR

4. a. Write a program to perform the following actions on a StringBuffer object: **08**
- **i.** Append a character sequence
- **ii.** Insert a string from a particular index to another index
- **iii.** Replace a string part from index 3 to 5 and put a string in that location.
- **iv.** Reverse a given string named "Hello World".

b. Write a program which use the following statements inside the try block: int a[]=new int[5]; a[5]=30/0; and handle the exception generated using 3 catch blocks with ArithmeticException, ArrayIndexOutOfBoundsException and Exception as class types. **06**

5. a. Compare Interfaces and classes in java with similarities and dissimilarities between them. **07**

b. "Java permits reference variables creation with interfaces but not objects." - Construct a program to support this claim. **07**

OR

6.	**a.**	Give 3 examples of built-in java packages. Write a program which shows static import.	**07**
	b.	Create 2 threads using a class which inherits the Runnable interface. Give a name to the thread. The threads would run a for loop and print its name and for loop iteration number.	**07**
7.	**a.**	Define Enumerator. Create a enum and show the use of values(), valueOf(), and ordinal() methods of Java enum.	**07**
	b.	What is generic functions? Write down a generic function which takes both integer and string input as argument.	**07**

OR

8.	**a.**	Construct a java program using InputStream and FileInputStream classes. Use the appropriate methods of these classes to achieve the following: **i.** Create a file named "input.txt" **ii.** Display Available bytes in the file **iii.** Data read from the file abd display in the console. **iv.** Close the stream.	**06**
	b.	Explain any four methods, which are widely used in Component class in java swing with example code.	**08**